

Quadratic equations



1 $x = 2$

2

a $a = -10$

b $a = 1$

3

a $m = 0, 14$

b $m = 5, -4$

c $m = 6, 1$

d $m = 6, 5$

e $m = 14, 13$

f $m = 0, 10$

g $m = 0, -2$

h $n = -4, -2$

i $y = 0, \frac{1}{3}$

j $y = 0, \frac{1}{5}$

k $y = 7, 1$

l $y = -1, 3$

m $b = -\frac{5}{3}, \frac{3}{4}$

n $k = 1, \frac{7}{6}$

4

a $x = \pm 8$

b $x = 0, -12$

c $x = -7, -6$

d $x = -7, 3$

e $x = 4, 10$

f $x = 10, -5$

g $x = 11$

h $x = 3, -2$

i $x = -8, -10$

j $x = 6, -8$

k $x = 2, -10$

l $x = 19, -6$

m $x = -\frac{5}{3}, 4$

n $x = -\frac{2}{5}, -4$

o $x = \frac{9}{4}, 4$

p $x = \frac{8}{5}, 9$

q $x = -\frac{1}{2}, 2$

r $x = \frac{3}{2}, 2$

s $x = \frac{5}{3}$

t $x = \frac{3}{5}, 2, \frac{5}{2}, 4$

5

a $x = 4, -7$

b $x = 8, -3$

c $x = 7, -4$

d $x = 9, -5$

e $x = -\frac{3}{2}, 3$

f $x = -\frac{5}{2}, 12$

g $x = -\frac{4}{3}, 5$

h $x = -7, 3$

i $x = 1, -\frac{7}{3}$

j $x = \frac{1}{2}, -\frac{5}{2}$

Applications

6 $x = \pm 4$

7

a $x = 2, -3$

b $x = 3, -7$

8

a $t = 3, 9$

b 3 hours

c 6 hours